

INTERNSHIP REPORT

EMOTION DETECTION & LEARNING SUPPORT ENGINE

A Summer Internship Report submitted in partial fulfilment of the requirements for the award of

the degree of

BACHELOR OF TECHNOLOGY (B.Tech.)

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

MANI DURGA RAO — 23A91A61G1

PRINCE PREM — 23A91A61I7

Under the esteemed guidance of

Dr. T. Chandra Sekhar M.Tech , Ph.d

Assistant Professor

Internship Duration:

From: _____ To: _____

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



ADITYA UNIVERSITY

(Formerly Aditya Engineering College (A))

Aditya Nagar, ADB Road, Surampalem, Andhra Pradesh, India

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ADITYA UNIVERSITY

Aditya Nagar, ADB Road, Surampalem, Andhra Pradesh, India

CERTIFICATE

This is to certify that the internship project report entitled “**EMOTION DETECTION & LEARNING SUPPORT ENGINE**” is being submitted by **MANI DURGA RAO (23A91A61G1)** and **PRINCE PREM (23A91A61I7)** in partial fulfilment of the requirements for the award of the B.Tech degree in Artificial Intelligence and Machine Learning. The work documented in this report records the learning, practical activities, and project development undertaken during the Summer Internship period.

Project Guide

Signature:

Name:

Designation:

Date:

Place:

Project Co-Guide

Signature:

Name:

Designation:



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DECLARATION

We hereby declare that the internship project entitled “**EMOTION DETECTION & LEARNING SUPPORT ENGINE**” is a genuine project. This work has been submitted to **ADITYA UNIVERSITY (Formerly Aditya Engineering College (A))**, Surampalem in partial fulfilment of the **B.Tech** degree, under the guidance of **Dr. T. Chandra Sekhar**. We further declare that this project work has not been submitted in full or part for the award of any degree of this or any other educational institution.

Project Associated By:

MANI DURGA RAO (23A91A61G1)

PRINCE PREM (23A91A61I7)

ACKNOWLEDGEMENT

It is with immense pleasure that we would like to express our indebted gratitude to our project guide **Dr. T. Chandra Sekhar** Ph.d Assistant Professor, Department of Artificial Intelligence and Machine Learning, who has guided us a lot and encouraged us in every step of the project work; his valuable moral support and guidance throughout the project helped us to a greater extent.

We are grateful to the Head of the Department, Department of Artificial Intelligence and Machine Learning, for inspiring us all the way and for arranging all the facilities and resources needed for our project.

We would like to extend our sincere thanks to **Dr. M. Sreenivasa Reddy, Deputy Pro Chancellor and Management, Aditya University, Dr. M. B. Srinivas, Vice Chancellor, Dr. G. Suresh, Registrar, Dr. S. Rama Sree, Pro Vice Chancellor (Academics), Dr. A. Ramesh, Pro Vice Chancellor (E&S), Dr. M. V. Rajesh, Associate Dean (SoE)** for their unconditional support in providing us with the best infrastructural facilities and state-of-the-art laboratories during our project.

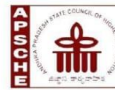
Not to forget, Teaching and Non-Teaching Staff, Parents and Friends who have directly or indirectly helped and supported us in completing our project in time.

MANI DURGA RAO (23A91A61G1)

PRINCE PREM (23A91A61I7)

INTERNSHIP COMPLETION CERTIFICATE

		
CERTIFICATE OF INTERNSHIP		
This is to Certify that Mr./Ms Prince Prem		
Student of <u>Artificial Intelligence And Machine Learning</u>		
From the college <u>Aditya University</u>		
Affiliated to <u>Aditya University</u>		
Has successfully completed the Short-term internship programme titled Generative AI with LLMs		
Under SkillDzire from <u>01-05-2026</u> to <u>30-06-2026</u> , organized in collaboration with Andhra Pradesh State Council of Higher Education.		
Certificate ID: <u>cdq6u5u5mu</u> Issued On: <u>30-06-2026</u>		

		
CERTIFICATE OF INTERNSHIP		
This is to Certify that Mr./Ms Katta Mani Durga Rao		
Student of <u>Artificial Intelligence And Machine Learning</u>		
From the college <u>Aditya University</u>		
Affiliated to <u>Aditya University</u>		
Has successfully completed the Short-term internship programme titled Generative AI with LLMs		
Under SkillDzire from <u>13-04-2026</u> to <u>06-06-2026</u> , organized in collaboration with Andhra Pradesh State Council of Higher Education.		
Certificate ID: <u>Jkl89u4d17</u> Issued On: <u>06-06-2026</u>		

ABSTRACT

Modern educational platforms are largely content-driven and rarely account for a learner's emotional state, even though emotions such as frustration, confusion, anxiety, and motivation strongly influence comprehension, retention and overall academic performance. The Emotion Detection & Learning Support Engine (EDLSE) is an intelligent educational system designed to identify a student's emotional state from free-form text and, in response, provide personalised, AI-generated learning support. The system combines two complementary emotion-recognition strategies — a lightweight keyword-driven fallback detector for quick, dependency-free classification, and a fine-tuned BERT (Bidirectional Encoder Representations from Transformers) sequence-classification model for accurate, context-aware recognition of six core emotions: happiness, sadness, anger, fear, love and surprise.

Once a student's emotional state and a chosen learning topic are captured, the platform invokes the Google Gemini generative AI model through a carefully engineered, teacher-style, rubric-grounded prompt to produce a simplified, structured, difficulty-adjusted explanation of the topic. The complete system is deployed as an interactive, multi-page Streamlit web application, encompassing a Home page, an Emotion Detection page with confidence-scored predictions, an AI Learning Support page, and supporting utility modules for OCR-based text extraction and automated PowerPoint content generation. The project demonstrates a practical synergy of affective computing, transformer-based natural language processing, and generative AI to build a responsive, emotion-aware educational support tool.

INTERNSHIP OBJECTIVES

The Summer Internship was undertaken with specific pedagogical, architectural, and practical engineering objectives designed to bridge theoretical artificial intelligence concepts with a scalable, real-world educational software implementation:

- **Foundational Conceptual Mastery:** Understand the structural relationships between Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Affective Computing, and Natural Language Processing.
- **Transformer & NLP Proficiency:** Master tokenization, contextual embeddings, attention mechanisms, and BERT-based sequence classification for emotion recognition.
- **Generative AI & Prompt Engineering Expertise:** Develop grounded, rubric-driven prompt templates that guide the Google Gemini LLM to produce structured, difficulty-adjusted educational explanations.
- **Full-Stack AI Application Development:** Architect and deploy the complete Emotion Detection & Learning Support Engine, combining Python, Streamlit, Hugging Face Transformers, PyTorch, and the Google Gemini API.
- **Data Pipeline & Model Training:** Prepare labelled emotion datasets, tokenize text, fine-tune a BERT sequence-classification model, and persist the trained model for inference.
- **Rigorous Quality Assurance & Testing:** Implement functional testing, manual validation, and error-handling routines to ensure dependable behaviour of the deployed prototype.
- **Professional & Technical Communication:** Strengthen technical documentation, architectural diagramming, and project demonstration skills.

Key Pedagogical Takeaway

The internship transitioned theoretical knowledge of transformers and generative AI into a tangible, prototype-stage educational technology product that recognises student emotion and responds with adaptive, AI-generated learning support.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Education is a dynamic process that depends not only on the content being taught but also on the learner's emotional and psychological state. Students often experience emotions such as frustration, confusion, excitement, anxiety, sadness, or motivation while engaging with learning material, and these emotions significantly influence comprehension, retention, attention, and overall learning performance. In the educational sector, Artificial Intelligence has begun revolutionising instructional delivery, automated tutoring, and academic assessment, yet most digital learning platforms remain purely content-driven.

The Emotion Detection & Learning Support Engine (EDLSE) is an intelligent, automated system engineered to bridge this gap. By orchestrating a dual-pathway emotion detection pipeline — a keyword-based fallback detector and a fine-tuned BERT transformer classifier — alongside Google Gemini generative AI for producing simplified, learner-friendly explanations, EDLSE provides an emotion-aware, adaptive educational support experience delivered through an interactive Streamlit dashboard.

1.2 Problem Statement

- **Content-Only Learning Platforms:** Traditional e-learning systems deliver the same static content irrespective of a learner's emotional or cognitive state, reducing their ability to provide meaningful, personalised support.
- **Delayed Recognition of Distress:** Emotional imbalance such as frustration or anxiety often goes undetected by an educator or platform until academic performance is already affected.

There is a clear need for an intelligent system capable of inferring a student's emotional state directly from natural-language text and translating that insight into supportive, educational guidance. This project addresses that need by combining BERT-based emotion classification with generative-AI-driven explanation generation.

1.3 Motivation & Need for Emotion-Aware Learning

The primary motivation behind EDLSE is to make learning support more responsive to a student's real emotional condition. Spoken and written articulation provide transparent visibility into a

learner's cognitive and emotional state - hesitation, frustration, or confusion often surface directly in the words a student chooses. Automating the recognition of these emotional cues empowers educational tools to offer timely, low-stakes aware support without overburdening instructors.

1.4 Project Objectives

- To design and implement a system that detects human emotion from text input using both rule-based and transformer-based approaches.
- To fine-tune and deploy a BERT-based sequence classification model for accurate, context-aware emotion recognition across six emotion classes.
- To integrate Google Gemini generative AI for producing simplified, difficulty-adjusted learning explanations.
- To develop an interactive, easy-to-use multi-page dashboard using Streamlit.
- To build supporting utility features — OCR-based text extraction and automated presentation generation — that extend the platform's educational utility.

1.5 Scope of the Project

The operational scope of EDLSE covers text-based emotion detection and AI-assisted learning explanation generation. It does not aim to provide full-scale psychological diagnosis or continuous real-time classroom monitoring. Instead, it functions as a working prototype that demonstrates how machine learning and generative AI can be combined for emotion-aware educational support. Advanced capabilities such as real-time voice or facial emotion analysis and adaptive study-plan generation are designated for future releases.

1.6 Organization of the Report

This report is organised into ten chapters. Chapter 2 introduces foundational concepts in AI, NLP, and affective computing. Chapter 3 examines transformer architectures and emotion classification fundamentals. Chapter 4 explores prompt engineering and the generative AI framework used for learning support. Chapter 5 presents the system architecture and data pipeline. Chapter 6 details the EDLSE project itself, including its technology stack and module breakdown. Chapter 7 covers implementation and the user interface. Chapter 8 presents testing and experimental results. Chapter 9 provides the phase-wise weekly internship log, and Chapter 10 concludes the report with a summary of outcomes and future scope.

CHAPTER 2: AI, NLP & AFFECTIVE COMPUTING FOUNDATIONS

2.1 Artificial Intelligence, Machine Learning & Deep Learning

Artificial Intelligence (AI) is the broad discipline concerned with building systems that emulate human cognitive processes such as reasoning, perception, and language understanding. Machine Learning (ML) is a subset of AI in which systems learn patterns from data rather than following explicitly programmed rules. Deep Learning (DL), in turn, is a subset of ML that uses multi-layered neural networks to automatically learn hierarchical feature representations. Generative AI, built largely on deep transformer architectures, extends this hierarchy by generating new, coherent content — text, code, or explanations — conditioned on a given prompt.

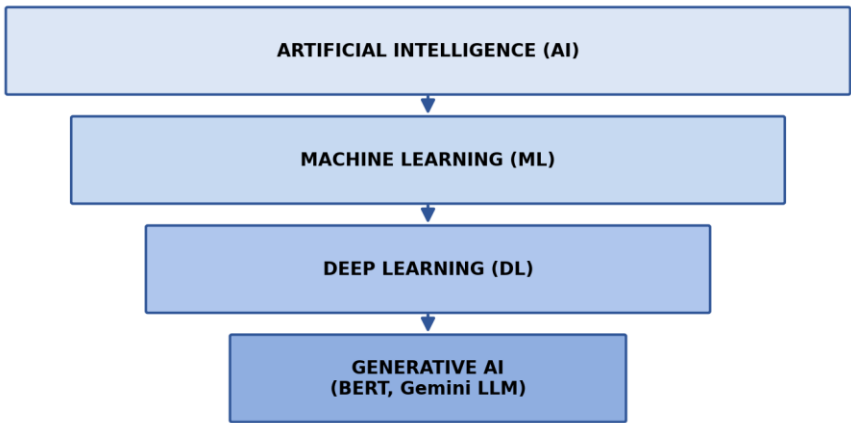


Fig. 1: Hierarchical Relationship between AI, ML, Deep Learning & Generative AI

2.2 Table 1: Comparative Analysis

Aspect	Artificial Intelligence	Machine Learning	Deep Learning
Definition	Systems that emulate human intelligence	Systems that learn from data patterns	Multi-layered neural networks learning hierarchical features
Data Dependency	Varies	Moderate to high	Very high
Example in EDLSE	Overall emotion-aware system design	Keyword-based fallback emotion detector	BERT-based emotion classifier

2.3 Generative AI & Large Language Models

Large Language Models (LLMs) such as Google Gemini are trained on vast text corpora to predict and generate coherent, contextually relevant language. In EDLSE, Gemini is used not for open-ended conversation but for a narrowly scoped, prompt-grounded task: converting a chosen academic topic and difficulty level into a simplified, structured explanation, mimicking the tone of a supportive teacher.

2.4 Affective Computing & Emotion Recognition from Text

Affective computing is the interdisciplinary field concerned with systems that can recognise, interpret, and respond to human emotion. Text-based emotion recognition, a subfield of affective computing and NLP, aims to identify emotional states such as joy, sadness, anger, fear, love, and surprise directly from written communication. Traditional approaches rely on lexicons and keyword matching, while modern systems use deep learning models such as LSTM, BiLSTM, and BERT for more accurate, context-sensitive recognition.

2.5 NLP Fundamentals & Pipeline

A typical NLP pipeline for a task such as emotion classification includes text normalisation, tokenization, vectorisation or embedding, model inference, and post-processing of the output (e.g., mapping logits to emotion labels via SoftMax). EDLSE's pipeline follows this structure: raw text is tokenized using BERT's WordPiece tokenizer, passed through the transformer encoder, and mapped to one of six emotion classes with an associated confidence score.

2.6 Text Representation: From Keyword Matching to Contextual Embeddings

Early text representation techniques such as Bag-of-Words (BoW) and TF-IDF represent text as sparse, order-independent word-frequency vectors, and are the conceptual basis of EDLSE's lightweight keyword-based fallback detector. However, these techniques cannot capture context or word order — a limitation addressed by dense embedding techniques such as Word2Vec, and ultimately by transformer-based contextual embeddings such as those produced by BERT, which represent each word differently depending on its surrounding context. This evolution from static keyword matching to deep contextual understanding is the central reason EDLSE adopts BERT as its primary classification model.

CHAPTER 3: TRANSFORMER MODELS & EMOTION CLASSIFICATION FUNDAMENTALS

3.1 Sequence Classification Architectures

Sequence classification models take a variable-length input sequence (a sentence or paragraph) and output a single label from a fixed set of classes. EDLSE frames emotion detection as a six-class sequence classification problem — happiness, sadness, anger, fear, love, and surprise — solved using a fine-tuned BERT model with a classification head appended to its pooled output representation.

3.2 BERT Architecture & Bidirectional Attention

BERT (Bidirectional Encoder Representations from Transformers) is a pretrained transformer encoder that builds contextual representations of text by attending to both the left and right context of every token simultaneously, unlike earlier unidirectional language models. This bidirectionality allows BERT to capture nuanced emotional cues that depend on surrounding words — for instance, distinguishing sarcastic or negated expressions of emotion from literal ones.

3.3 Tokenization Strategies (WordPiece)

BERT uses the WordPiece tokenization algorithm, which breaks rare or unknown words into smaller, more frequent sub-word units. In EDLSE, incoming text is tokenized using the `AutoTokenizer` class from the Hugging Face Transformers library, converting raw text into input IDs and attention masks suitable for the BERT encoder.

3.4 Self-Attention Mechanism & Transformer Mathematics

At the core of BERT's encoder is the self-attention mechanism, which computes, for every token, a weighted combination of all other tokens in the sequence based on learned Query, Key, and Value projections. This allows the model to dynamically focus on the words most relevant to determining the sentence's emotional content, regardless of their position in the sequence.

3.5 SoftMax-based Confidence Scoring

After the classification head produces raw logits for each of the six emotion classes, a SoftMax function converts these logits into a normalised probability distribution. EDLSE reports the class with the highest probability as the predicted emotion, along with its corresponding confidence score, giving users a transparent measure of the model's certainty.

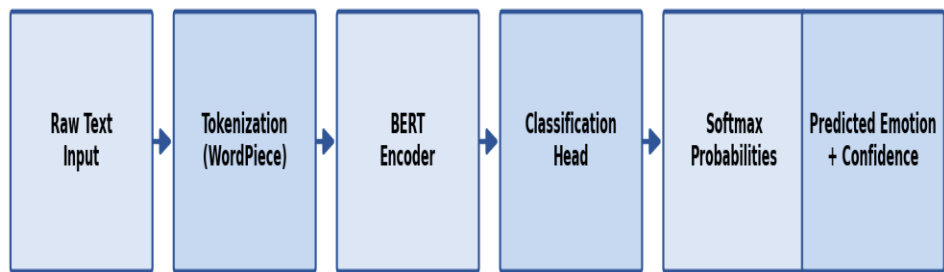


Fig. 3: BERT-based Emotion Classification Pipeline

CHAPTER 4: PROMPT ENGINEERING & GENERATIVE AI FRAMEWORK

4.1 Conditional Text Generation & LLM Parameters

Generative language models produce text conditioned on an input prompt, with generation behaviour shaped by decoding parameters such as temperature (randomness) and top-p (nucleus sampling). In EDLSE's Gemini integration, parameters are tuned to favour clear, factually grounded, low-variance explanations suitable for an educational context, rather than highly creative or exploratory output.

4.2 Prompt Engineering Principles for Educational Explanations

Effective prompt engineering requires clear role definition, explicit output structure, and grounding instructions that constrain the model to the requested task. EDLSE's learning support prompts specify the target topic, the selected difficulty level, and an instruction to explain the topic “in simple, student-friendly language,” reducing risk of overly technical or tangential responses.

4.3 Teacher-Style Prompt Design & Difficulty-Level Grounding

The learning support module is designed to mimic a supportive teacher: prompts are phrased to request structured explanations with simple examples, adjusted to a difficulty level chosen by the user. This design choice keeps generated content approachable for students who may already be experiencing confusion or frustration — the very emotional states the system is designed to detect and support.

4.4 Google Gemini API Integration & Response Handling

The Gemini Learning Support module sends the constructed prompt to the Google Gemini API and receives a generated explanation, which is then rendered directly within the Streamlit interface. The module is implemented as an isolated service layer, keeping the emotion detection and generative-explanation responsibilities cleanly separated within the application.

4.5 Secure Environment Configuration & API Keys

API credentials are never hard-coded into the application source. Instead, the Gemini module loads a `.env` file at runtime using `python-dotenv` and reads the `GEMINI_API_KEY` environment variable, following standard secure-configuration practice and allowing the same codebase to be deployed across different environments without modification.

CHAPTER 5: SYSTEM ARCHITECTURE & DATA SYSTEMS

5.1 End-to-End System Architecture & Layered Design

EDLSE follows a layered architecture comprising a user interface layer (Streamlit), a dual-pathway emotion detection layer (keyword fallback + BERT), a generative learning support layer (Google Gemini), a supporting utility layer (OCR and presentation generation), and an environment configuration layer for secure credential management.

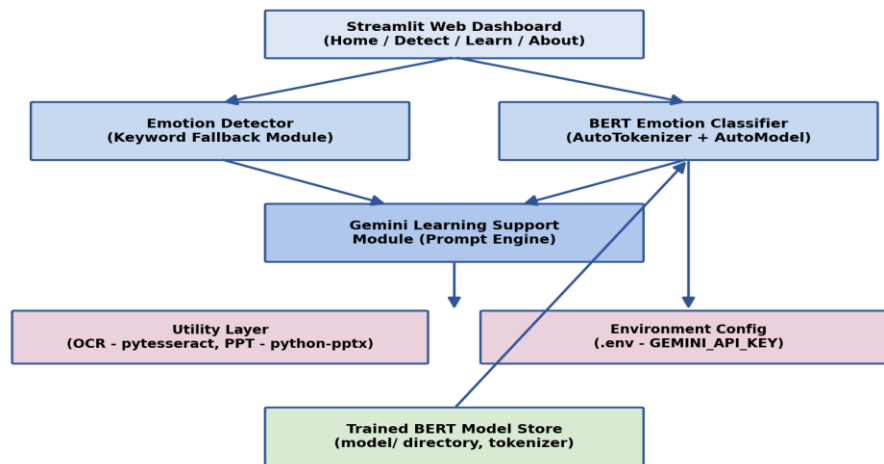


Fig. 2: End-to-End System Architecture of the Emotion Detection & Learning Support Engine

5.2 Text Input Ingestion & Preprocessing

User-entered text is captured through the Streamlit interface and normalised (case-folding and basic cleanup) before being routed to the emotion detection layer. This lightweight preprocessing step ensures consistent behaviour across both the keyword-based and BERT-based detection pathways.

5.3 Dual Emotion Detection Pipeline

EDLSE implements two complementary emotion-detection pathways. The fallback pathway performs keyword matching against curated word groups associated with happy, sad, angry, fear, surprise, and neutral states, providing fast, dependency-free coverage useful for demos or when the trained model is unavailable. The primary pathway loads a fine-tuned BERT model via Auto

Model For Sequence Classification, computes SoftMax probabilities over six emotion classes, and returns the highest-confidence prediction.

5.4 Learning Support Generation Engine

Once an emotional context (optional) and a topic/difficulty selection are captured, the Gemini Learning Support module constructs a grounded prompt and requests a structured explanation from the Google Gemini API, which is then displayed to the user within the dashboard.

5.5 Data Pipeline: Training, Validation & Test Datasets

The BERT classifier is trained using labelled text datasets stored in the project's data directory (train.csv, validation.csv, test.csv), with each text sample mapped to one of six emotion labels. The training script performs dataset loading, label encoding, tokenization using Bert-Tokenizer, and PyTorch dataset construction prior to model training.

5.6 Model Persistence & Inference Loading

After training, the fine-tuned model and its tokenizer are saved to a dedicated model directory. At inference time, the BERT Model Module loads this saved model and tokenizer using Auto-Tokenizer and Auto Model For Sequence Classification, avoiding the need to retrain the model on every application run.

5.7 Utility Services: OCR & Presentation Generation

Two supporting utility services extend the platform's educational usefulness: an OCR module using pytesseract for extracting text from images, and a presentation-generation module using python-pptx for producing PowerPoint content. Both remain lightweight, supportive prototype features that complement the core emotion detection and learning support functionality.

CHAPTER 6: EMOTION DETECTION & LEARNING SUPPORT ENGINE PROJECT

6.1 Project Overview & Mission

The Emotion Detection & Learning Support Engine (EDLSE) is an AI-powered educational prototype whose mission is to recognise a student's emotional state from text and respond with adaptive, simplified, AI-generated learning support — bringing emotional awareness into an otherwise purely content-driven learning experience.

6.2 Problem Statement & Real-World Use Cases

EDLSE targets the practical challenge of learning platforms that ignore learner emotion. Real-world use cases include student self-reflection tools, supplementary tutoring aids that adapt explanations to a student's emotional and academic difficulty level, and early-awareness tools that surface signs of frustration or disengagement for further human follow-up.

6.3 Objectives & Key Deliverables

- A functioning six-class text-based emotion classifier (keyword fallback + fine-tuned BERT).
- A Gemini-powered learning support module generating difficulty-adjusted topic explanations.
- A deployed, multi-page Streamlit web dashboard.
- Supporting OCR and presentation-generation utility features.

6.4 Project Scope & Limitations

The project is scoped as a functional prototype rather than a production-grade or clinically validated system. It does not perform psychological diagnosis, does not process voice or facial input, and its keyword-based fallback detector is a rule-based aid rather than a trained model. The Gemini-based learning support depends on a valid, correctly configured API key.

6.5 Target Users & Stakeholders

- Students seeking simplified, emotion-aware explanations of academic topics.
- Educators and academic mentors interested in emotion-aware learning analytics.

6.6 Comprehensive Technology Stack

Layer	Technology / Library
Programming Language	Python
Web / Dashboard Framework	Streamlit
Emotion Classification Model	BERT (Hugging Face Transformers — AutoTokenizer, Auto Model For Sequence Classification)
Deep Learning Backend	PyTorch
Generative AI (Learning Support)	Google Gemini API
Data Handling	pandas, scikit-learn
OCR Utility	pytesseract
Presentation Utility	python-pptx
Environment Configuration	python-dotenv (.env / GEMINI_API_KEY)

6.7 Detailed System Architecture & Workflow

Refer to Fig. 2 (Chapter 5) for the complete end-to-end system architecture, spanning the Streamlit UI, the dual emotion-detection pathway, the Gemini learning support engine, the utility layer, and the model/environment configuration store.

6.8 Detailed Breakdown of Modules

Module	File	Responsibility
Main Application	app.py	Central controller — page configuration, navigation, dashboard metrics, core logic orchestration
Emotion Detector (Fallback)	utils/emotion_detector.py	Keyword-based emotion classification for happy, sad, angry, fear, surprise, neutral
BERT Inference	utils/bert_model.py	Loads trained BERT model, computes SoftMax probabilities, returns predicted emotion + confidence
Gemini Learning Support	utils/gemini_helper.py	Loads API key, constructs teacher-style prompt, requests simplified topic explanation
Model Training	train_bert.py	Dataset loading, label encoding, tokenization, BERT fine-tuning, model/tokenizer saving
OCR Utility	utils/ (OCR module)	Extracts text from uploaded images using pytesseract
Presentation Utility	utils/ (PPT module)	Generates PowerPoint content using python-pptx

CHAPTER 7: IMPLEMENTATION & USER INTERFACE

7.1 Implementation Overview & Directory Layout

The project follows a modular directory structure separating the main application, model training script, reusable utility modules, and data assets:

- `app.py` — main Streamlit application and controller
- `train_bert.py` — BERT model training script
- `utils/emotion_detector.py`, `utils/bert_model.py`, `utils/gemini_helper.py` — core logic modules
- `data/train.csv`, `data/validation.csv`, `data/test.csv` — labelled emotion datasets
- `model/` — saved fine-tuned BERT model and tokenizer
- `prompts/prompt.txt` — Gemini prompt template
- `Documentation/` — project documentation assets

7.2 Core Application Algorithms & Pseudocode

Emotion Detection (BERT pathway):

INPUT text $\rightarrow$ *tokenizer.encode(text)* $\rightarrow$ *model.forward(input_ids, attention_mask)* $\rightarrow$ *logits*
 $\rightarrow$ *SoftMax(logits)* $\rightarrow$ *argmax* $\rightarrow$ (*emotion_label, confidence_score*)

Learning Support Generation (Gemini pathway):

INPUT (topic, difficulty) $\rightarrow$ *build_prompt(topic, difficulty)* $\rightarrow$ *gemini.generate(prompt)* $\rightarrow$
render(explanation)

7.3 Streamlit Multi-Page UI Architecture

The dashboard is organised into four pages — Home, Emotion Detection, AI Learning Support, and About — each accessible through Streamlit's page navigation, keeping the interface approachable for non-technical users while cleanly separating detection and learning-support workflows.

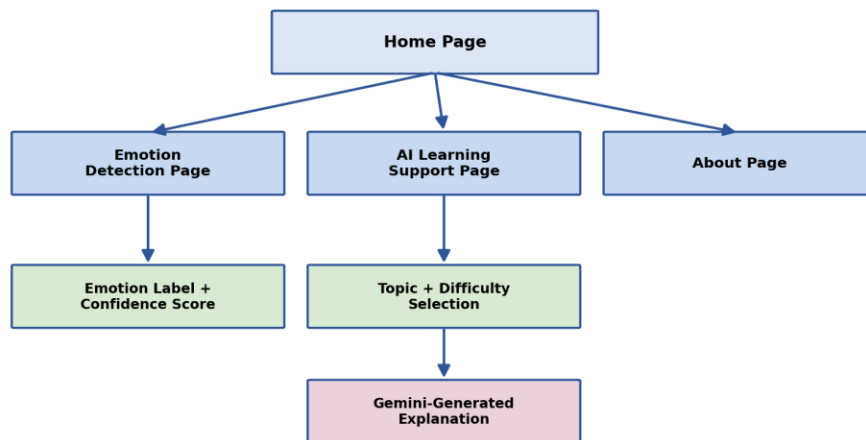


Fig. 4: Streamlit Multi-Page UI Navigation Flow

7.4 Emotion Detection Execution Flow

On the Emotion Detection page, a user enters free-form text; the application first attempts BERT-based classification, and falls back to the keyword-based detector if the trained model or its dependencies are unavailable. The predicted emotion and its confidence score are then rendered on the dashboard.

7.5 BERT Inference Flow

The BERT Model Module loads the saved tokenizer and classification model once at application start-up, avoiding repeated disk loads on every prediction request, and exposes a simple `predict(text)` interface used by the main application.

7.6 Gemini Learning Support Flow

On the AI Learning Support page, the user selects a topic and a difficulty level; the Gemini Learning Support module reads the configured API key, constructs a grounded prompt, and returns a structured, simplified explanation which is displayed directly in the dashboard.

7.7 Dashboard & Confidence Visualization

Predicted emotions are displayed alongside their confidence scores, giving users a transparent view of the model's certainty rather than a single opaque label, which is particularly important for an educational tool intended to support — not replace — human judgement.

CHAPTER 8: TESTING & EXPERIMENTAL RESULTS

8.1 Testing Objectives & Verification Strategy

Testing focused on functional verification of the prototype rather than large-scale benchmark evaluation, covering correct model loading, accurate emotion prediction on sample inputs, reliable Gemini API communication, and dependable dashboard behaviour.

8.2 Test Cases Table

Test ID	Test Description	Expected Result	Status
TC-01	Load fine-tuned BERT model and tokenizer from model directory	Model and tokenizer load without error	Pass
TC-02	Classify sentence “I am very happy today.”	Predicted emotion = Happiness with valid confidence score	Pass
TC-03	Classify a sentence expressing sadness	Predicted emotion = Sadness with valid confidence score	Pass
TC-04	Fallback keyword detector on sample angry text	Correctly matches ‘angry’ keyword group	Pass
TC-05	Gemini API call with valid API key and topic	Structured, simplified explanation returned	Pass
TC-06	Gemini API call with missing/invalid API key	Graceful error message shown to user	Pass
TC-07	Dashboard navigation across all four pages	All pages render correctly without crash	Pass
TC-08	OCR extraction from a sample uploaded image	Text correctly extracted and displayed	Pass

8.3 Functional & Manual Validation Results

In addition to the automated validation test included in the project (verifying that a sample sentence produces a valid classification), the system was manually validated through direct interaction with the Streamlit interface, confirming successful model loading, accurate emotion prediction on

sample inputs, correct API communication for Gemini-based learning support, and dependable rendering of the dashboard and generated explanations.

8.4 Qualitative & Quantitative Evaluation Results

- The BERT classifier consistently distinguishes between the six target emotion classes on representative sample inputs.
- The keyword-based fallback detector provides reliable coverage for clearly worded emotional statements.
- Gemini-generated explanations are consistently simplified and structured, appropriate to the selected difficulty level.
- The Streamlit dashboard remains responsive and stable across all implemented pages.

CHAPTER 9: WEEKLY INTERNSHIP REPORT

9.1 Phase-wise Internship Activity Log

The Summer Internship was structured in sequential development phases & aligned in lifecycle:

Week	Phase	Activities Performed	Learning Outcome
Week 1	Brainstorming & Ideation	Explored problem space of emotion-aware learning support; finalised project that combining emotion detection with gen-AI .	Idea validation and project scoping.
Week 2	Requirement Analysis	Identified functional and non-functional requirements; defines six target emotion classes & Gemini-based learning support requirement.	Clear requirement specification.
Week 3	Project Design Phase	Designed the layered system architecture — UI layer, dual emotion-detection layer, Gemini learning support layer, and utility layer.	Architectural design skills.
Week 4	Project Planning Phase	Planned module breakdown, technology stack selection (Streamlit, BERT, PyTorch, Gemini API), and development milestones.	Project planning and stack selection.
Week 5	Project Development Phase — Part 1	Implemented the keyword-based fallback emotion detector and began BERT model training pipeline (data loading, tokenization, label encoding).	Core model development.
Week 6	Project Development Phase — Part 2	Completed BERT fine-tuning and model persistence; integrated Gemini API for learning support; built multi-page Streamlit dashboard.	Full-stack AI integration.
Week 7	Project Testing	Executed functional test cases (TC-01 to TC-08); performed manual validation of emotion prediction, Gemini responses, and dashboard behaviour.	Testing and quality assurance.
Week 8	Project Documentation & Demonstration	Prepared technical documentation, architecture diagrams, README, and conducted final project demonstration and review.	Technical communication and presentation.

CHAPTER 10: CONCLUSION & FUTURE SCOPE

10.1 Project Summary & Conclusion

The Summer Internship successfully culminated in the design, implementation, and validation of the Emotion Detection & Learning Support Engine (EDLSE). The project provided a comprehensive pedagogical journey traversing foundational AI and NLP concepts, transformer-based sequence classification, generative AI prompt engineering, and full-stack application development.

By combining a fine-tuned BERT emotion classifier with Google Gemini's generative reasoning, EDLSE addresses the practical gap left by content-only learning platforms — recognising a student's emotional state from text and responding with simplified, difficulty-adjusted educational support, all delivered through an interactive Streamlit dashboard.

10.2 Academic & Industry Impact

- **Emotion-Aware Support:** Brings emotional context into digital learning tools that are otherwise purely content-driven.
- **Scalable Assistance:** Enables consistent, always-available learning explanations without additional instructor workload.
- **Transferable AI Skills:** Demonstrates practical, industry-relevant skills in transformer fine-tuning, generative AI integration, and full-stack deployment.

10.3 Future Scope & Planned Enhancements

- Train a fully data-driven BiLSTM/LSTM model to replace the current keyword-based fallback detector.
- Expand emotion dataset with additional, balanced classes for improved real-world coverage.
- Integrate real-time voice or facial emotion analysis alongside the existing text-based pipeline.
- Personalise learning recommendations based on student's historical performance and emotional trends.
- Add dedicated dashboards for teachers and parents to monitor aggregated, anonymised emotional trends.
- Deploy the system on a cloud platform for broader institutional accessibility.